

Research Note: Rethinking Low-Level Design (LLD) Practice & Evaluation

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1. Executive Summary & Problem Analysis

Low-Level Design (LLD) and Object-Oriented Domain Modeling are core components of senior software engineering assessments. However, while **Data Structures & Algorithms (DSA)** benefits from deterministic automated test fixtures (e.g. LeetCode, HackerRank) and **System Design (HLD)** benefits from high-level capacity calculations and block architecture diagrams, **Low-Level Design suffers from an acute evaluation gap.**

When a learner designs a *Parking Lot*, *Elevator Controller*, or *Rate Limiter*, they face fundamental uncertainty:

Are my class responsibilities cohesive (Single Responsibility Principle)?

Did I over-engineer with unnecessary design patterns, or did I hardcode logic that violates the Open-Closed Principle?

Is my concurrency handling sound, or will it deadlock under multi-threaded gate access?

How does my solution compare to alternative valid architectural paradigms?

Without targeted, explainable, and iterative feedback, learners default to passive memorization of static GitHub repositories, which fails to develop real engineering judgement.

2. Competitive Landscape & Existing Approaches Researched

We analyzed four prevailing methods candidates use to practice LLD:

- Requirements & Clarifying Assumptions** (Defining scope, actors, and constraints)
- Class Structure / Model** (Entities, attributes, methods, and relationships)
- Design Pattern Rationale** (Justifying why specific GoF patterns were chosen)
- Core Implementation Code** (Demonstrating interfaces, encapsulation, and type safety)
- Trade-offs & Concurrency** (Addressing thread-safety, race conditions, and OCP extension points)

B. Standardized 6-Dimension Evaluation Rubric

We evaluate all submissions against an objective, 100-point dimensional rubric:

$$\text{Total Score} = \sum_{i=1}^6 (\text{Score}_i \times \text{Weight}_i)$$

Each criterion strictly adheres to the schema:

$$\text{Criterion} \rightarrow \text{Score} \rightarrow \text{Evidence} \rightarrow \text{Concern} \rightarrow \text{Suggestion} \rightarrow \text{Confidence}$$

C. First-Class Multi-Attempt Delta Analytics

The platform retains attempt history, enabling learners to re-attempt problems and instantly visualize dimensional score shifts (ΔScore), verifying that previous architectural weaknesses were systematically resolved.

5. Conclusion

By separating deterministic structural checks from rubric-driven evaluation, providing evidence-backed suggestions, and tracking iterative progress, the LLD Practice Platform transforms Low-Level Design from a subjective guessing game into a rigorous, learnable engineering discipline.